

Python Intermediate Day 2 Checkpoint

Object-Oriented Programming Fundamentals

- OOP – The Big Picture
- Class Coding Basics
- Realistic OOP Example
- Class Coding Details

1. Email *

2. Name

3. **What best describes the difference between a class and an instance?** 1 point

Mark only one oval.

- ☐ A. A class is data, an instance is code
- ☐ B. A class is a blueprint, an instance is a created object
- ☐ C. A class exists at runtime, an instance exists at compile time
- ☐ D. There is no difference

4. **Which concept allows a subclass to provide its own implementation of a method?** 1 point

Mark only one oval.

- ☐ A. Encapsulation
- ☐ B. Inheritance
- ☐ C. Polymorphism
- ☐ D. Composition

5. **Which method is automatically called when an object is created?** 1 point

Mark only one oval.

- ☐ A. new
- ☐ B. start
- ☐ C. init
- ☐ D. create

6. **What does the self keyword represent inside a class method?** 1 point

Mark only one oval.

- ☐ A. The class itself
- ☐ B. The current instance
- ☐ C. A global variable
- ☐ D. A parent class

7. **Which method should be overridden to provide a readable string for users?**

1 point

Mark only one oval.

- ☐ A. repr
- ☐ B. debug
- ☐ C. str
- ☐ D. print

8. **What is the main purpose of inheritance?**

1 point

Mark only one oval.

- ☐ A. To reduce memory usage
- ☐ B. To reuse and extend existing code
- ☐ C. To hide data
- ☐ D. To prevent modification

9. **What does super() do?**

1 point

Mark only one oval.

- ☐ A. Calls a static method
- ☐ B. Refers to the current object
- ☐ C. Calls methods from the parent class
- ☐ D. Creates a new instance

10. Which tool allows inspection of object attributes at runtime?

1 point

Mark only one oval.

- ☐ A. print()
- ☐ B. help()
- ☐ C. introspection (e.g., dir(), hasattr())
- ☐ D. compile()

11. What is an Abstract Base Class mainly used for?

1 point

Mark only one oval.

- ☐ A. To store data
- ☐ B. To enforce method implementation in subclasses
- ☐ C. To improve performance
- ☐ D. To replace interfaces

12. Which module is commonly used for object persistence in Python?

1 point

Mark only one oval.

- ☐ A. json
- ☐ B. csv
- ☐ C. pickle
- ☐ D. sys

13. A lambda function in Python is:

1 point

Mark only one oval.

- ☐ A. A multi-line function
- ☐ B. An anonymous, single-expression function
- ☐ C. A replacement for classes
- ☐ D. Used only in AI

14. What is encapsulation in OOP?

1 point

Mark only one oval.

- ☐ A. Combining multiple classes
- ☐ B. Hiding internal details and exposing behavior
- ☐ C. Using inheritance
- ☐ D. Overloading operators

15. What is the main difference between `__str__` and `__repr__`?

1 point

Mark only one oval.

- ☐ A. repr is slower
- ☐ B. str is for users, repr is for developers/debugging
- ☐ C. str replaces print()
- ☐ D. They behave exactly the same

16. **Why is this code useful?**

1 point

```
if hasattr(acc, "apply_interest"): acc.apply_interest()
```

Mark only one oval.

- ☐ A. Prevents syntax errors
- ☐ B. Ensures method exists before calling it
- ☐ C. Improves performance
- ☐ D. Makes code shorter

17. **Which is a good AI development best practice?**

1 point

Mark only one oval.

- ☐ A. Trust model output blindly
- ☐ B. Use one AI model for all tasks
- ☐ C. Choose models based on strengths and use cases
- ☐ D. Avoid documentation

18. **What is the purpose of using an @ mention in GitHub Copilot Chat?**

1 point

Mark only one oval.

- ☐ A. To tag another developer in your code
- ☐ B. To direct Copilot to use a specific context, tool, or participant when responding
- ☐ C. To create a code comment
- ☐ D. To mark a syntax error

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